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The Editor
Physical Review B
American Physical Society

Dear Editor,

I am pleased to resubmit the following revised manuscript for consideration in *Physical Review B*:

"Special Optimal Thermal Cell: Correlator-Designed Supercells for Effective-Harmonic Phonons and Thermodynamics"

Albert Linda

We thank the reviewers for identifying several important issues in the original submission. The revision substantially narrows the claims and adds direct, auditable controls. SOTC is now presented as deterministic configuration design followed by symmetry-constrained effective second-order force-constant (FC2) fitting. It is not presented as a variational anharmonic theory or as a general replacement for TDEP, SSCHA, or molecular dynamics.

The revised manuscript reports nine metal and rocksalt benchmarks. Their archived workflows reach the scalar correlator stopping criterion in 2–5 iterations using 4–36 force calculations, including reference structures; these counts are reported as measurements rather than as a universal speedup. A new matched Quantum ESPRESSO comparison for Al, W, and NaCl evaluates 20 canonical-Gaussian sequences and three pre-specified correlator-matched sequences at identical force-call counts. At 16 calls the nominal SOTC dispersion errors differ from the canonical medians by +3.2%, +8.1%, and −2.8%, respectively. We therefore conclude only that the nominal design is comparable in these tests and explicitly report its initialization sensitivity.

The revision also corrects the factor-of-three mean-square-displacement convention, removes the underconverged quasi-harmonic calculation and all dependent claims, replaces unsupported rank statements with auditable force-call and saved-matrix diagnostics, and defines reference-force subtraction explicitly. New calculations provide non-analytic LO–TO corrections for NaCl, MgO, and PbTe; tabulated 80 K NaCl neutron data are shown directly; and the Au instability is quantified and excluded from the stable dispersion evidence. The response letter gives a point-by-point account of these changes.

The revised source, response letter, validation scripts, compact machine-readable evidence, and generated figures are available in the public SOTC repository.

The manuscript is a good fit for *Physical Review B*: it presents a first-principles method for condensed-matter thermodynamics, supported by theoretical derivations, systematic validation, and open reproducibility.

This manuscript has not been submitted elsewhere and is not under consideration at any other journal. I have no conflicts of interest to declare.

Thank you for your time and consideration.

Sincerely,

Albert Linda